

Braid Presentations for Concurrent Programs

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Abstract

Ziemiański’s cube chain category $\text{Ch}(K)$ is a combinatorial model for the space of directed paths on a precubical set K . Paliga and Ziemiański built a braid invariant for precubical sets by observing a relation between $\pi_1|\text{Ch}(Z)|$ and braids, where Z is the terminal precubical set. However, that invariant trivializes for Euclidean K because $|\text{Ch}(I^n)|$ is contractible [PZ21, Prop. 6.1]. This paper constructs a localization of $\text{Ch}(K)$ at morphisms that preserve the order of coordinates, called $\text{Br}(K)$. And we will show that this structure is the source of the braid structure of $\pi_1|\text{Ch}(Z)|$. But, there is no nerve or geometric realization in this technique. Instead we build a 2-polygraph giving an artin braid-like presentation of $\text{Br}(K)$. The $\{0, 1, 2\}$ -cells of the polygraph are the dimension $\{0, 1, 2\}$ chains of K . Intuitively, 3-cells of K contribute braid relations, and pairs of non-intersecting 2-cells of K contribute commuting relations. This is not just a metaphor: at $K = Z$, we get the artin relations and generators exactly. So $\text{Br}(Z)$ is equivalent to the positive braid category. The presentation is natural in K , providing a generalization of the existing work. As an application, we establish a link to the hyperplane arrangement literature. The way braids are constructed via Salvetti’s theorem for the braid arrangement exactly matches the polygraph we build here for the cube. All this work is formalized in Lean [Sto26].

1 Introduction

► *Background:* $\text{Ch}(K)$ as a model of executions [Zie20]; the Paliga–Ziemiański invariant and why it trivializes on Euclidean complexes [PZ21, Prop. 6.1].

► *The move:* an object of $\text{Ch}(K)$ cuts an execution of N events into consecutive blocks of simultaneous ones, a morphism relabels the events and coarsens the blocking. Say which refinements are order preserving, and that inverting them is a localization and not a quotient, so the arrows of $\text{Br}(K)$ still record which events crossed which and how often.

► *Why the generators and relations are not chosen:* they are the chains of degree 1 and 2, and a chain is already arrow-shaped, from $\text{dom } \perp_e$ to $\text{dom } \top_e$.

Theorem 1.1 (Theorem 5.5, Theorem 6.1, Theorem 7.6). *Let K be a bipointed precubical set, $\text{Br}(K)$ the localization of $\text{Ch}(K)^{\text{op}}$ at the order-preserving refinements (Definition 3.5), and $\text{Art}(K)$ the 2-polygraph of Definition 4.14, whose 0-cells are the runs of K and whose 1- and 2-cells are its chains of degree 1 and 2.*

1. *There is a functor $E_K : \langle \text{Art}(K) \rangle \rightarrow \text{Br}(K)$, sending a 1-cell α to $Q(\top_\alpha^{\text{op}}) \circ Q(\perp_\alpha^{\text{op}})^{-1}$, and it is an equivalence.*

2. $K \mapsto \text{Art}(K)$ extends to a functor $\square_* \rightarrow \mathbf{Poly}$, and for every $f : K \rightarrow L$,

$$E_L \circ \langle \text{Art}(f) \rangle = \text{Br}(f) \circ E_K.$$

3. $\text{Art}(Z) \cong \mathcal{A}$ in $\mathbf{Poly}$, where $\mathcal{A}$ is Artin's braid polygraph (Definition 7.5). Consequently $\text{Br}(Z) \simeq \mathbf{Braid}^{+, \text{op}}$.

None of the three assumes anything about K .

► Say why $\text{Art}(K)$ deserves the name “Artin polygraph” before (iii) is proved: its 2-cells already split into the two species of Proposition 4.13, a hexagon at a block of dimension 3 and a square at two blocks of dimension 2, which are Artin's braid and commutation relations. Then note that (iii) makes this literal.

► Relation to Gaussent–Guiraud–Malbos [GGM15]: the same 2-cells, there from Garside theory and here from the geometry of K , with the Artin case the value at a point.

Formalisation. All of the below is formalised in Lean 4 on mathlib [Sto26], sorry-free. The objects $\text{Br}(K)$, W and $\text{Art}(K)$ are `ChainCat.Br`, `ChainCat.W` and `Paper.poly`; the three parts of Theorem 1.1 are `Paper.paperPresents`, `Paper.paperSquare` and `Paper.paperArtinIso`. Each statement below appears in `CubeChains/Paper.lean` as an `example` discharged by one citation.

Conventions. **Cube** is the box category, $\square = \widehat{\mathbf{Cube}}$ the precubical sets, $\square_*$ the bipointed ones with Z final, and I^n the n -cube, bipointed by $(0, \dots, 0)$ and $(1, \dots, 1)$. We set $[n] := \{1, \dots, n\}$ and identify $K_n = \text{Hom}_{\square}(I^n, K)$ by Yoneda. Composition is classical: gf means “first f , then g ”. A word in a polygraph is a path read left to right, so $\sigma_i \sigma_k$ is the arrow $\sigma_k \circ \sigma_i$.

► Explain the two $^{\text{op}}$ s: refinements run from finer to coarser [PZ21; Zie20], and words read left to right, so $\text{Br}(K)$ localizes $\text{Ch}(K)^{\text{op}}$ and $\langle \mathcal{A} \rangle$ is $\mathbf{Braid}^{+, \text{op}}$.

2 Background

► One paragraph placing Proposition 2.2: everything downstream is bookkeeping about the events of a wedge, and this is the only place the precubical structure is touched.

Cube chains [PZ21; Zie20]. The **wedge** $P \vee Q$ of $P, Q \in \square_*$ is the pushout in $\square$ gluing fin_P to init_Q , bipointed by init_P and fin_Q ; $(\square_*, \vee, I^0)$ is monoidal. A **shape** is a list $a = [a_1, \dots, a_r]$ of positive integers, with **serial wedge** $\vee a := I^{a_1} \vee \dots \vee I^{a_r}$ and **blocks** the I^{a_i} ; put $|a| := \sum_i a_i$ and $\deg a := \sum_i (a_i - 1) = |a| - r$. Then $\text{Ch}(K)$ is the full subcategory of $\square_*/K$ spanned by the serial wedges: an object is a bipointed map $\vee a \rightarrow K$ and a morphism, a **refinement**, is a map of wedges commuting over K . We write $\deg d$ and $|d|$ for the degree and event count of a chain's shape and $\text{Ch}(u) : \text{Ch}(K) \rightarrow \text{Ch}(L)$ for the pushforward along $u : K \rightarrow L$, which postcomposes the classifying map and leaves the shape alone. A chain of degree 0 is a **run**; the full subcategory $\text{Run}(K)$ of runs is discrete, a refinement losing no block being an identity.

We use three facts from [PZ21, §2]: serial wedges are non-self-linked and admit an altitude function $\text{alt} : K_0 \rightarrow \mathbb{Z}$ with $\text{alt}(\text{fin } e) = \text{alt}(\text{init } e) + 1$ on edges, the final vertex of $\vee a$ having altitude $|a|$; a bipointed map between connected objects admitting altitude preserves it [PZ21, Prop. 2.7]; and

$\text{Ch}(P) \times \text{Ch}(Q) \rightarrow \text{Ch}(P \vee Q)$, concatenation, is an isomorphism when $P \vee Q$ is non-self-linked and admits altitude [PZ21, Lem. 2.11].

The events of a shape. Let $\text{Coord} : \mathbf{Cube} \rightarrow \mathbf{Set}$ send I^n to $[n]$ and a face embedding g to the monotone injection g_* naming the directions g leaves free, and let $E(\text{Coord})$ be its left Kan extension along Yoneda. That functor is cocontinuous and $\text{Coord}(I^0) = \emptyset$, so the pushout defining a wedge becomes a coproduct:

$$\text{Ev}(a) := E(\text{Coord})\left(\bigvee a\right) \cong \coprod_i [a_i].$$

Write $f_* := E(\text{Coord})(f)$.

Convention 2.1. We identify $\text{Ev}(a)$ with $[N]$, $N = |a|$, by listing $\coprod_i [a_i]$ block by block, and write $\text{blk}_a : [N] \rightarrow [r]$ for the resulting block map. A shape on N events is thus a partition of $[N]$ into consecutive intervals, and it enters everything below only through blk_a . We drop the subscript when the shape is clear.

Proposition 2.2. *Let $f : \bigvee a \rightarrow \bigvee b$ in $\square_*$, with a and b shapes on N and M events. Then $N = M$; the map f_* is a permutation of $[N]$, and $f \mapsto f_*$ is injective; and there is a monotone $\text{blk}(f)$ with*

$$\text{blk}_b \circ f_* = \text{blk}(f) \circ \text{blk}_a, \quad (1)$$

the restriction of f_ to each block of a being increasing.*

Proof. By the pushout, f is determined by its restrictions to the blocks of a , each of which lands in a single block of b because $a_i > 0$; that gives $\text{blk}(f)$, equation (1) and the last clause, and a face embedding is determined by the directions it leaves free, so f_* determines f . Injectivity of f_* and monotonicity of $\text{blk}(f)$ follow by induction on the length of b : by [PZ21, Lem. 2.11] a map $\bigvee a \rightarrow I^m \vee \bigvee b'$ splits as $f_1 \vee f_2$ over a splitting of a , so f_* is a disjoint union and $\text{blk}(f)$ is $\text{blk}(f_1) \sqcup (1 + \text{blk}(f_2))$ with $\text{blk}(f_1)$ constant at 1; injectivity of the left half is a second induction, the two images at each glued vertex being separated by the coordinates already equal to 1 there. Finally f preserves altitude [PZ21, Prop. 2.7], so $N = M$ and the injection f_* is a permutation. $\square$

Presented categories. A **2-polygraph** $\mathcal{P}$ consists of a set V of 0-cells, families $\mathcal{P}_1(x, y)$ and $\mathcal{P}_2(x, y)$ for $x, y \in V$, and two maps $\text{src}, \text{tgt} : \mathcal{P}_2(x, y) \rightarrow \{\text{paths } x \rightarrow y\}$ into the free category on the 1-cells; $\text{src}(\alpha)$ and $\text{tgt}(\alpha)$ are the **words** of α . A morphism of polygraphs is a map in each dimension commuting with src and tgt on the nose, and **Poly** is the resulting category. The category $\langle \mathcal{P} \rangle$ **presented by** $\mathcal{P}$ is the free category on the 1-cells modulo the congruence generated by $\text{src}(\alpha) = \text{tgt}(\alpha)$, with quotient functor q .

Proposition 2.3 ([Ara+23, Ch. 2]). *Let $\mathcal{P}$ be a 2-polygraph and C a category.*

1. *A functor $\langle \mathcal{P} \rangle \rightarrow C$ is the same thing as a map $V \rightarrow \text{Ob } C$ together with an arrow of C for each 1-cell, such that the two words of every 2-cell are sent to the same arrow.*
2. *Two functors out of $\langle \mathcal{P} \rangle$ that agree on 0-cells and 1-cells are equal.*
3. *A morphism of polygraphs bijective in every dimension is an isomorphism.*

Proposition 2.4 ([GZ67, Ch. I]). *Let W be a class of morphisms of a category C . There is a functor $Q : C \rightarrow C[W^{-1}]$ inverting W through which every functor inverting W factors uniquely, and two functors out of $C[W^{-1}]$ that agree after Q are equal.*

► *One sentence: the only arrows of $\text{Br}(K)$ written down below are composites $Q(g^{\text{op}})Q(m^{\text{op}})^{-1}$ coming from a cospan whose first leg is order preserving, and every identity used about them is an instance of Q being a functor.*

Newman's lemma. Write $\text{Hasse}(P)$ for the Hasse quiver of a poset P . A **reading** of P in a category C assigns an object to each element and an arrow to each covering relation; it extends to paths by composition.

Theorem 2.5 (coherent Newman's lemma; [Hue80], [Ara+23, Ch. 4]). *Let P be a finite poset and φ a reading of P in C . Suppose that for every pair of distinct covers $x \triangleleft y_1$, $x \triangleleft y_2$ and every common upper bound z of y_1 and y_2 there are $j \leq z$ and paths p_i from y_i to j with $\varphi(x \triangleleft y_1) \varphi(p_1) = \varphi(x \triangleleft y_2) \varphi(p_2)$. Then φ sends parallel paths to equal arrows, so it extends uniquely to a functor $P \rightarrow C$; and a functor out of P is determined by its values on covers.*

Lemma 2.6. *Let P be a finite poset, graded by r with $r(y) = r(x) + 1$ on covers, in which every element has exactly two Hasse neighbours. Then a directed path is determined by its first edge; and if P has a least and a greatest element, it has exactly two maximal chains, of a common length ℓ , and $|P| = 2\ell$.*

Proof. Undirected, the Hasse graph is 2-regular, hence a disjoint union of cycles, and r increases along a directed path, so a path leaves each vertex by the edge it did not arrive on. Both edges at a least element point up and both at a greatest point down, so the cycle through the least element meets the greatest one and splits into two directed arcs, of equal length by the grading; that cycle is all of P , since every element lies on a path from the least to the greatest. $\square$

3 Squares, crossings, and the braiding

► *One paragraph: a square is a pair of events of one block, and the whole construction runs on the question of which squares a map reverses.*

Throughout, a and b are shapes on N events, read as partitions of $[N]$ into consecutive intervals (Convention 2.1).

Definition 3.1. A **square** of a is a pair $s = (p, q)$ of events with $p < q$ and $\text{blk}(p) = \text{blk}(q)$; write $\text{Sq}(a)$ for the set of them, so $|\text{Sq}(a)| = \sum_i \binom{a_i}{2}$. A set $R \subseteq \text{Sq}(a)$ is **biclosed** when for all events $p < q < r$ of one block,

$$(p, q) \in R \text{ and } (q, r) \in R \implies (p, r) \in R, \quad (p, r) \in R \implies (p, q) \in R \text{ or } (q, r) \in R.$$

We write $\text{Sq}(d)$ for the squares of a chain's shape, and f_* also for the induced injection $\text{Sq}(a) \rightarrow \text{Sq}(b)$, $(p, q) \mapsto (f_*p, f_*q)$, well defined because f_* is increasing on blocks (Proposition 2.2).

The second clause says that the complement of R satisfies the first, so the biclosed sets are closed under complementation.

Definition 3.2. A map of serial wedges $f : \bigvee a \rightarrow \bigvee b$ **crosses**

$$\text{Cr}(f) := \{ (p, q) \in \text{Sq}(b) : \text{blk}_a(f_*^{-1}q) < \text{blk}_a(f_*^{-1}p) \},$$

the squares of b whose second event f brings from a strictly earlier block of a than its first.

Lemma 3.3. *A square (p, q) of b lies in the image of f_* if and only if $f_*^{-1}p$ and $f_*^{-1}q$ lie in one block of a . For events u, v of b in distinct blocks, $\text{blk}_a(f_*^{-1}u) < \text{blk}_a(f_*^{-1}v)$ if and only if $\text{blk}_b(u) < \text{blk}_b(v)$.*

Proof. A square in the image pulls back to a square, whose events share a block; conversely if the two preimages share a block then, f_* being increasing there, they form a square mapping to (p, q) . The second claim is (1) with $\text{blk}(f)$ monotone, read in both directions, the two blocks being distinct. $\square$

Lemma 3.4. *For $f : \bigvee a \rightarrow \bigvee b$ and $g : \bigvee b \rightarrow \bigvee c$,*

$$\text{Cr}(gf) = g_*(\text{Cr } f) \sqcup \text{Cr}(g),$$

a disjoint union, and $\text{Cr}(\text{id}) = \emptyset$.

Proof. Let s be a square of c . If $s = g_*(t)$ then $s \notin \text{Cr}(g)$, the two events of t sharing a block (Lemma 3.3) — which also gives disjointness — and $(gf)_*^{-1}$ agrees with f_*^{-1} on the events of t , so $s \in \text{Cr}(gf)$ if and only if $t \in \text{Cr}(f)$. If s is not in the image of g_* then it is not in $g_*(\text{Cr } f)$, and its two events pull back along g_* into distinct blocks of b , so the second part of Lemma 3.3 applied to f gives $s \in \text{Cr}(gf)$ if and only if $s \in \text{Cr}(g)$. $\square$

Definition 3.5. A refinement f is **order preserving** when $\text{Cr}(f) = \emptyset$. Write $W = W_K$ for the class of order-preserving refinements of $\text{Ch}(K)$, and

$$\text{Br}(K) := \text{Ch}(K)^{\text{op}}[(W^{\text{op}})^{-1}], \quad [c] := Q(c).$$

Proposition 3.6. *A refinement f is order preserving if and only if $f_* = \text{id}_{[N]}$. Hence W contains the identities; $gf \in W$ if and only if $f \in W$ and $g \in W$; any two order-preserving refinements with the same source and target are equal; and $\text{Ch}(u)$ carries W_K into W_L , so it descends to $\text{Br}(u) : \text{Br}(K) \rightarrow \text{Br}(L)$, strictly functorially in u .*

Proof. If $f_* = \text{id}$ and (p, q) is a square of b , then $\text{blk}(f)(\text{blk}_a p) = \text{blk}_b(p) = \text{blk}_b(q) = \text{blk}(f)(\text{blk}_a q)$ by (1), while $p < q$ and blk_a monotone give $\text{blk}_a(p) \leq \text{blk}_a(q)$; so $(p, q) \notin \text{Cr}(f)$.

Conversely let $\text{Cr}(f) = \emptyset$ and $p < q$ in $[N]$. If $\text{blk}_a(p) = \text{blk}_a(q)$ then $f_*p < f_*q$, as f_* is increasing on blocks. If $\text{blk}_a(p) < \text{blk}_a(q)$ then $\text{blk}_b(f_*p) \leq \text{blk}_b(f_*q)$ by (1), so $f_*q < f_*p$ would force both into one block of b , making (f_*q, f_*p) a square crossed by f . So f_* is monotone, hence the identity.

The consequences follow: the second from Lemma 3.4 with g_* injective, the third because f_* determines f (Proposition 2.2), and the last because $\text{Ch}(u)$ changes neither shape nor wedge map, with the descent given by Proposition 2.4. $\square$

► *One or two sentences reading Proposition 3.6 back: an order-preserving refinement is one that only glues adjacent blocks, the elementary ones being $\bigvee[\dots, p, q, \dots] \rightarrow \bigvee[\dots, p + q, \dots]$.*

4 The Artin polygraph of K

► *Frame the section: the runs over a chain form a poset; its covers are the degree-one chains over that chain, and the rank-two intervals are the degree-two ones. That is the whole polygraph.*

Fix K , and let d be a chain of K on N events.

4.1 Runs over a chain

Definition 4.1. A **run over** d is a refinement $x : X \rightarrow d$ whose source X is a run; write $\mathcal{R}(d)$ for the set of these. Since X has shape $[1, \dots, 1]$, the permutation x_* enumerates the events of d in the order x fires them; write $p <_x q$ when $x_*^{-1}(p) < x_*^{-1}(q)$, the **firing order** of x .

A refinement out of a run is determined by its map of wedges, the classifying map of the source being forced; so $\mathcal{R}(d)$ is the set of runs of the wedge $\bigvee a$, and $\text{Cr}(x)$ is the set of squares (p, q) of d with $q <_x p$.

Lemma 4.2. *A run over d fires the blocks of d in order: $\text{blk}(p) < \text{blk}(q)$ implies $p <_x q$. So x is a choice of linear order on each block of d , and blk is a monotone surjection from the firing order onto the blocks, with interval fibres.*

Proof. The blocks of the source are singletons, so $\text{blk} \circ x_* = \text{blk}(x)$, monotone by Proposition 2.2. A monotone surjection out of a chain has interval fibres, here the blocks of d ; the only freedom left is the order inside each. $\square$

Theorem 4.3. $x \mapsto \text{Cr}(x)$ is a bijection from $\mathcal{R}(d)$ onto the biclosed subsets of $\text{Sq}(d)$, and reversing the order on every block complements Cr .

Proof. Both the statement and biclosedness are local to a block, by Lemma 4.2; fix one and let $<$ be a linear order on it. Then $\text{Cr}(<) = \{(p, q) : p < q, q < p\}$ is its inversion set, and $<$ is recovered from a set R by declaring $p < q$ when either $p < q$ and $(p, q) \notin R$, or $q < p$ and $(q, p) \in R$. So the map is injective, and its image is the set of R for which that recipe is transitive, totality being automatic. Transitivity can fail only on a triple $p < q < r$, where the cases to check are the two biclosedness clauses. $\square$

► *Note that for a single block this is the classical description of inversion sets, and that inclusion of them is the right weak order [BB05, §3.1]; the order theory below is that statement, block by block.*

We order $\mathcal{R}(d)$ by $x \leq y : \iff \text{Cr}(x) \subseteq \text{Cr}(y)$, and write $\perp_d, \top_d$ for its least and greatest elements, which exist because $\emptyset$ and $\text{Sq}(d)$ are biclosed.

Proposition 4.4. *Let $q : o \rightarrow d$ be a refinement. Composition with q is an order embedding $\mathcal{R}(o) \rightarrow \mathcal{R}(d)$, and*

$$\text{Cr}(qx) = q_*(\text{Cr}(x)) \sqcup \text{Cr}(q).$$

Moreover $q\perp_o = \perp_d$ if and only if $q \in W$; for $x \in \mathcal{R}(d)$ the conditions $x \in W$, $\text{Cr}(x) = \emptyset$ and $x = \perp_d$ agree; $\perp_X = \text{id}_X$ for a run X ; and $\text{dom } \perp_d = \text{dom } \perp_c$ whenever $c \rightarrow d$ lies in W .

Proof. The formula is Lemma 3.4. Since q_* is injective with image disjoint from $\text{Cr}(q)$ (Lemma 3.3), $qx \leq qy$ if and only if $x \leq y$, and $\text{Cr}(q\perp_o) = \text{Cr}(q)$. The remaining equivalences are Theorem 4.3, and the last clause is the formula applied to $c \rightarrow d$. $\square$

4.2 Steps, covers, and cells

Lemma 4.5. *Call a square s a **step** of $x \in \mathcal{R}(d)$ when $\text{Cr}(x) \triangle \{s\}$ is biclosed, write $\text{St}(x)$ for the set of steps and x^s for the run with $\text{Cr}(x^s) = \text{Cr}(x) \triangle \{s\}$. Then s is a step of x if and only if its two events are consecutive in the firing order of x , and $|\text{St}(x)| = \deg d$.*

Proof. A block's events are consecutive in the firing order (Lemma 4.2), so “consecutive in the block” and “consecutive overall” agree. If the two events of s are consecutive, toggling s swaps an adjacent pair of a linear order, so the result is again a linear order and is biclosed by Theorem 4.3. If a third event of the block fires strictly between them, toggling s swaps the pair but not the order of either event against that third one, which would then have to lie both before and after the pair.

So the steps are the covers of the firing order lying inside a block. The firing order is a chain on N elements and blk is a monotone surjection from it onto the r blocks (Lemma 4.2); such a map collapses exactly $N - r = \deg d$ covers. $\square$

Lemma 4.6. *If $x < z$ in $\mathcal{R}(d)$ then some $s \in \text{St}(x) \setminus \text{Cr}(x)$ lies in $\text{Cr}(z)$. Hence $x < y$ if and only if $\text{Cr}(y) = \text{Cr}(x) \cup \{s\}$ for such an s , in which case $y = x^s$; and $\mathcal{R}(d)$ is graded by $|\text{Cr}|$, from 0 at $\perp_d$ to $|\text{Sq}(d)|$ at $\top_d$.*

Proof. If z agreed with x on every pair covering for $<_x$, then $<_x$ would be contained in $<_z$ by transitivity, and two linear orders so related are equal, so $z = x$. So some covering pair $p <_x q$ has $q <_z p$. Its two events share a block, since otherwise Lemma 4.2 would make every run fire p first, so the pair is a step s of x (Lemma 4.5). If $s = (p, q)$ then $s \in \text{Cr}(z) \setminus \text{Cr}(x)$, as wanted; if $s = (q, p)$ then $s \in \text{Cr}(x) \subseteq \text{Cr}(z)$, which says $p <_z q$, a contradiction.

Given such an s we have $x \leq x^s \leq y$, forcing $y = x^s$; conversely two biclosed sets differing by one element are a covering pair, by Theorem 4.3. $\square$

Proposition 4.7. *Let $x \in \mathcal{R}(d)$. Sending a factorisation $X \rightarrow m \rightarrow d$ of x to the set of steps of x whose two events share a block of m is a bijection onto the subsets of $\text{St}(x)$; write m_S for the middle at S and $\iota_S : X \rightarrow m_S$, $\pi_S : m_S \rightarrow d$ for its legs. Then $\deg m_S = |S|$, and if $S \cap \text{Cr}(x) = \emptyset$ — we then say S **ris**es from x — then ι_S is order preserving.*

Proof. A middle m makes the N events into its blocks, which by Lemma 4.2 applied to ι are intervals of the firing order of x , and π requires each such interval to lie inside one block of d ; conversely such an interval partition arises from the chain whose blocks are its parts, in order. The factorisation is determined by that shape: a map of wedges is determined by its action on events and injective there (Proposition 2.2), hence a monomorphism, while the first leg is forced by which events fall in which part. An interval partition of a chain is the same thing as the set of covers it does not cut; those covers lie inside a block, so they are steps (Lemma 4.5); and $\deg m$ is N minus the number of parts, which is the number of uncut covers.

Let S rise from x . Since ι_S is a run over m_S , its crossings are the squares of m_S that x reverses, that is the pairs inside one part that x reverses. The covers of the firing order inside a part all lie in S , hence none is reversed, so the firing order agrees with the event order on consecutive pairs of the part and hence, by transitivity, on every pair of it. So $\text{Cr}(\iota_S) = \emptyset$. $\square$

Definition 4.8. For runs $X, Y \in \text{Run}(K)$, an n -**cell** from X to Y is a chain e with $\deg e = n$, $\text{dom } \perp_e = X$ and $\text{dom } \top_e = Y$; write $\text{Cell}_n(X, Y)$ for the set of them.

► *One sentence: so the n -cells are the chains of degree n , sorted by the two runs each spans, and a 0-cell is a run.*

Lemma 4.9. *Let S rise from $x \in \mathcal{R}(d)$. Then*

$$\text{Cr}(\pi_S \top_{m_S}) = \text{Cr}(x) \cup \pi_{S*}(\text{Sq}(m_S)),$$

and every $z \geq x$ with $S \subseteq \text{Cr}(z)$ satisfies $\pi_{S}(\text{Sq}(m_S)) \subseteq \text{Cr}(z)$. So $\pi_S \top_{m_S}$ is the least element of $\mathcal{R}(d)$ above x reversing all of S .*

Proof. Proposition 4.4 gives $\text{Cr}(\pi_S \top_{m_S}) = \pi_{S*} \text{Sq}(m_S) \cup \text{Cr}(\pi_S)$, and Lemma 3.4 applied to $\pi_S \iota_S = x$ with $\text{Cr}(\iota_S) = \emptyset$ gives $\text{Cr}(\pi_S) = \text{Cr}(x)$.

For the second claim list a part as $p_1 < \dots < p_k$; as in the previous proof this is also its firing order, so each (p_i, p_{i+1}) lies in S and is reversed by z . Since $\text{Cr}(z)$ is biclosed, the first clause applied to the triples (p_i, p_{i+1}, p_j) gives, by induction on $j - i$, that z reverses every pair of the part. $\square$

Corollary 4.10. *Let $x \leq y$ in $\mathcal{R}(d)$, with step s . Then $\gamma(x \leq y) := m_{\{s\}}$ is a 1-cell from $\text{dom } x$ to $\text{dom } y$, with $\pi_{\{s\}} \perp_{m_{\{s\}}} = x$ and $\pi_{\{s\}} \top_{m_{\{s\}}} = y$; and every 1-cell α is γ of the unique cover $\perp_\alpha \leq \top_\alpha$ of $\mathcal{R}(\alpha)$. So γ , together with $x \mapsto \text{dom } x$ on vertices, is a map of quivers from $\text{Hase}(\mathcal{R}(d))$ to the quiver of 1-cells, and $\gamma(qe) = \gamma(e)$ for every refinement $q : o \rightarrow d$ and every cover e of $\mathcal{R}(o)$.*

Proof. $m_{\{s\}}$ has degree 1 and $\text{dom } \perp_{m_{\{s\}}} = \text{dom } x$ by Proposition 4.7; by Lemma 4.9 its top composes forward to the least element above x reversing s , namely $x^s = y$. Conversely a chain α of degree 1 has one square, so $\mathcal{R}(\alpha)$ is the two-element poset $\perp_\alpha \leq \top_\alpha$, and the factorisation through its one step is α . For the last claim the step of qe is q_* of the step of e (Proposition 4.4), and q_* carries the firing order of x to that of qx , so the two factorisations cut the same partition and have the same middle. $\square$

4.3 The attachment of a 2-cell

Lemma 4.11. *Order $\text{Sq}(d)$ lexicographically, by the smaller event and then the larger. Every lower set of that order is biclosed, and so is every upper set. Consequently, for $\prec$ either that order or its reverse, the runs x_t with $\text{Cr}(x_t)$ the first t squares form a maximal chain $\perp_d = x_0 \prec \dots \prec x_{|\text{Sq}(d)|} = \top_d$, in which the $(t+1)$ -st square is a step of x_t .*

Proof. On a triple $p < q < r$ of one block the three squares come in the order $(p, q) \prec (p, r) \prec (q, r)$. For a lower set R , membership of (q, r) forces that of (p, r) , and membership of (p, r) forces that of (p, q) , which are the two clauses; for an upper set, $(p, q) \in R$ forces $(p, r) \in R$ and $(p, r) \in R$ forces $(q, r) \in R$. The chain then exists by Theorem 4.3, consecutive terms differ by one element, and Lemma 4.6 applies. $\square$

Definition 4.12. For a chain e with $\deg e = 2$, let $\text{src}(e)$ and $\text{tgt}(e)$ be the words of 1-cells read off under γ from the two chains of Lemma 4.11, for the lexicographic order and for its reverse.

Proposition 4.13. *Let $\deg e = 2$. Then $\mathcal{R}(e)$ is a polygon on $2|\text{Sq}(e)|$ elements whose only two maximal chains from $\perp_e$ to $\top_e$ are those of Lemma 4.11; so $\text{src}(e)$ and $\text{tgt}(e)$ have length $|\text{Sq}(e)|$ and differ in their first letter. Moreover either e has one block of dimension 3, $|\text{Sq}(e)| = 3$ and $\mathcal{R}(e)$ is a hexagon, or e has two blocks of dimension 2, $|\text{Sq}(e)| = 2$ and $\mathcal{R}(e)$ is a square.*

Proof. Every $x \in \mathcal{R}(e)$ has exactly two steps (Lemma 4.5), and the two flips are its two Hasse neighbours, upward or downward according to whether the step is already reversed (Lemma 4.6). The two chains of Lemma 4.11 leave $\perp_e$ by different covers, the least and greatest squares being

distinct, so Lemma 2.6 applies with the grading $|\text{Cr}|$. Finally $\deg e = 2$ forces the block dimensions above 1 to be $\{3\}$ or $\{2, 2\}$, and Definition 3.1 counts $\binom{3}{2} = 3$ and $2\binom{2}{2} = 2$. $\square$

Definition 4.14. The **Artin polygraph** $\text{Art}(K)$ has 0-cells the runs of K , 1-cells $\text{Art}(K)_1(X, Y) := \text{Cell}_1(X, Y)$, 2-cells $\text{Art}(K)_2(X, Y) := \text{Cell}_2(X, Y)$, and the words src , tgt of Definition 4.12.

► *One paragraph on the name: $\text{Art}(K)$ is built from K alone, with no braid theory in it, and Proposition 4.13 already exhibits its 2-cells as braid and commutation relations; Theorem 7.6 is what makes the name literal.*

Proposition 4.15. *Read in $\langle \text{Art}(K) \rangle$, the quiver map γ satisfies the hypothesis of Theorem 2.5 on $\mathcal{R}(d)$. So it extends uniquely to a functor $\Gamma_d : \mathcal{R}(d) \rightarrow \langle \text{Art}(K) \rangle$, and $\Gamma_d(qx) = \Gamma_o(x)$ for every refinement $q : o \rightarrow d$ and every $x \in \mathcal{R}(o)$.*

Proof. Let $x < y_1$, $x < y_2$ be distinct covers with steps $s_1 \neq s_2$, and z a common upper bound. Then $S = \{s_1, s_2\}$ rises from x , so $\alpha := m_S$ is a chain of degree 2, that is a 2-cell (Proposition 4.7), and $j := \pi_S \top_\alpha$ satisfies $j \leq z$ (Lemma 4.9). By Proposition 4.13 the two maximal chains of $\mathcal{R}(\alpha)$ are $\text{src}(\alpha)$ and $\text{tgt}(\alpha)$, and they leave $\perp_\alpha$ by its two covers. Composition with π_S carries $\perp_\alpha$ to x and $\top_\alpha$ to j , carries covers to covers (Proposition 4.4, Lemma 4.6) and keeps the 1-cell each cover reads as (Corollary 4.10). So the two chains become two paths from x to j , leaving by y_1 and by y_2 and reading as $\text{src}(\alpha)$ and $\text{tgt}(\alpha)$, which are equal in $\langle \text{Art}(K) \rangle$.

For the last claim, composition with q is monotone, so $x \mapsto \Gamma_d(qx)$ and Γ_o are functors out of $\mathcal{R}(o)$; they agree on covers by Corollary 4.10, hence agree, by the last clause of Theorem 2.5. $\square$

5 The presentation

► *Frame the proof as a comparison of two universal properties: a functor out of $\langle \text{Art}(K) \rangle$ is a reading of the degree-one chains identifying the two words of every degree-two chain (Proposition 2.3), and a functor out of $\text{Br}(K)$ is a functor on $\text{Ch}(K)^{\text{op}}$ inverting W (Proposition 2.4). Each is built from the other.*

A cell $\alpha \in \text{Cell}_n(X, Y)$ comes with the two refinements $\perp_\alpha : X \rightarrow \alpha$ and $\top_\alpha : Y \rightarrow \alpha$, of which the first is order preserving; since Q inverts $\perp_\alpha^{\text{op}}$, we may set

$$E_K(X) := [X], \quad E_K(\alpha) := Q(\top_\alpha^{\text{op}}) \circ Q(\perp_\alpha^{\text{op}})^{-1} : [X] \rightarrow [Y].$$

Lemma 5.1. *Let d be a chain and $z \in \mathcal{R}(d)$. Every path from $\perp_d$ to z in $\text{Hasse}(\mathcal{R}(d))$ reads, under $E_K \circ \gamma$, as $Q(z^{\text{op}}) \circ Q(\perp_d^{\text{op}})^{-1}$.*

Proof. Induct on the length of the path, the empty path reading as the identity. For a cover $x < y$ with step s , put $\beta := \gamma(x < y)$ and $q := \pi_{\{s\}}$, so $x = q\perp_\beta$ and $y = q\top_\beta$ (Corollary 4.10) and $x^{\text{op}} = \perp_\beta^{\text{op}} q^{\text{op}}$. The middle two factors of

$$E_K(\beta) \circ Q(x^{\text{op}}) \circ Q(\perp_d^{\text{op}})^{-1} = Q(\top_\beta^{\text{op}}) \circ Q(\perp_\beta^{\text{op}})^{-1} \circ Q(\perp_\beta^{\text{op}}) \circ Q(q^{\text{op}}) \circ Q(\perp_d^{\text{op}})^{-1}$$

cancel, leaving $Q(\top_\beta^{\text{op}})Q(q^{\text{op}})Q(\perp_d^{\text{op}})^{-1} = Q(y^{\text{op}})Q(\perp_d^{\text{op}})^{-1}$. $\square$

Corollary 5.2. *The two words of a 2-cell have the same image under $E_K \circ \gamma$, so E_K extends to a functor $E_K : \langle \text{Art}(K) \rangle \rightarrow \text{Br}(K)$.*

Proof. Both words are paths from $\perp_e$ to $\top_e$ (Proposition 4.13), so both read as $Q(\top_e^{\text{op}})Q(\perp_e^{\text{op}})^{-1}$ (Lemma 5.1); apply Proposition 2.3(i). $\square$

Definition 5.3. Let $\Theta_K : \text{Ch}(K)^{\text{op}} \rightarrow \langle \text{Art}(K) \rangle$ be given by $\Theta_K(d) := \text{dom } \perp_d$ and, for a refinement $u : c \rightarrow d$,

$$\Theta_K(u^{\text{op}}) := \Gamma_d(\perp_d \leq u\perp_c) : \text{dom } \perp_d \rightarrow \text{dom } \perp_c,$$

which typechecks because $u\perp_c \in \mathcal{R}(d)$ has source $\text{dom } \perp_c$.

Lemma 5.4. Θ_K is a functor, and $\Theta_K(u^{\text{op}})$ is an identity whenever u is order preserving; so Θ_K descends uniquely to $\Phi_K : \text{Br}(K) \rightarrow \langle \text{Art}(K) \rangle$ with $\Phi_K Q = \Theta_K$.

Proof. Let $v : b \rightarrow c$ and $u : c \rightarrow d$. Composition with u is monotone (Proposition 4.4), so $\perp_d \leq u\perp_c \leq uv\perp_b$, and Γ_d being a functor, $\Theta_K((uv)^{\text{op}}) = \Gamma_d(u\perp_c \leq uv\perp_b) \circ \Theta_K(u^{\text{op}})$; the left factor is $\Gamma_c(\perp_c \leq v\perp_b) = \Theta_K(v^{\text{op}})$ by the last clause of Proposition 4.15. If u is order preserving then $u\perp_c = \perp_d$ (Proposition 4.4), so the comparison is an identity. The descent is Proposition 2.4. $\square$

Theorem 5.5. For every bipointed precubical set K , the functor E_K is an equivalence $\langle \text{Art}(K) \rangle \rightarrow \text{Br}(K)$, with Φ_K inverse to it.

Proof. On a 0-cell, $\Phi_K(E_K(X)) = \Theta_K(X) = \text{dom } \perp_X = X$ (Proposition 4.4). On a 1-cell α , applying Φ_K to the defining composite gives $\Phi_K(E_K(\alpha)) = \Theta_K(\top_\alpha^{\text{op}}) \circ \Theta_K(\perp_\alpha^{\text{op}})^{-1}$, whose second factor is an identity because $\perp_\alpha$ is order preserving (Lemma 5.4). Since $\top_\alpha \perp_Y = \top_\alpha$ and $\mathcal{R}(\alpha)$ is the two-element poset $\perp_\alpha < \top_\alpha$,

$$\Theta_K(\top_\alpha^{\text{op}}) = \Gamma_\alpha(\perp_\alpha \leq \top_\alpha) = q(\gamma(\perp_\alpha < \top_\alpha)) = q(\alpha)$$

by Corollary 4.10. So $\Phi_K E_K$ agrees with the identity on 0- and 1-cells, hence is the identity (Proposition 2.3(ii)).

For the other composite, Lemma 5.1 gives $E_K(\Theta_K(u^{\text{op}})) = Q((u\perp_c)^{\text{op}})Q(\perp_d^{\text{op}})^{-1} = Q(\perp_c^{\text{op}}) \circ Q(u^{\text{op}}) \circ Q(\perp_d^{\text{op}})^{-1}$, which is $Q(u^{\text{op}})$ conjugated by the isomorphisms $[c] \cong [\text{dom } \perp_c]$ that Q makes of the order-preserving refinements $\perp_c$. So $E_K \Theta_K \cong Q$, and $E_K \Phi_K \cong \text{id}$ by uniqueness of descent (Proposition 2.4). $\square$

► *One sentence: the comparison is an equivalence and not an isomorphism because $\text{Br}(K)$ is not skeletal — a chain and the run below it become isomorphic there without being equal — so the runs are a skeleton of $\text{Br}(K)$.*

6 Naturality

Theorem 6.1. Sending K to $\text{Art}(K)$, and $f : K \rightarrow L$ to $\text{Ch}(f)$ on cells in every dimension, is a functor $\square_* \rightarrow \text{Poly}$; and for every $f : K \rightarrow L$,

$$E_L \circ \langle \text{Art}(f) \rangle = \text{Br}(f) \circ E_K : \langle \text{Art}(K) \rangle \longrightarrow \text{Br}(L).$$

Proof. Write $f_\bullet := \text{Ch}(f)$. A run over a chain e is determined by its map of wedges (Definition 4.1), and $f_\bullet e$ has the same shape and the same maps of wedges into it, so $\mathcal{R}(e) = \mathcal{R}(f_\bullet e)$ as ordered sets. Everything in Definition 3.2, Lemma 4.5 and Corollary 4.10 is defined from the map of wedges

alone, so $f_\bullet$ preserves $\perp$, $\top$, cells, steps, covers and γ ; and it carries the lexicographic order on $\text{Sq}(e)$ to the lexicographic order on $\text{Sq}(f_\bullet e) = \text{Sq}(e)$, so it preserves src and tgt letter by letter. Hence $\text{Art}(f)$ is a morphism of polygraphs, functorially in f because Ch is a functor.

Both composites send a run X to $[f_\bullet X]$. On a 1-cell α , $\text{Br}(f)$ is the descent of $Q_L \circ f_\bullet^{\text{op}}$, so it sends $E_K(\alpha)$ to $Q((f_\bullet \top_\alpha)^{\text{op}}) \circ Q((f_\bullet \perp_\alpha)^{\text{op}})^{-1}$. Here $f_\bullet \perp_\alpha$ is an order-preserving refinement into $f_\bullet \alpha$ out of a run, hence is $\perp_{f_\bullet \alpha}$ (Proposition 4.4), and likewise $f_\bullet \top_\alpha = \top_{f_\bullet \alpha}$. So both composites send α to $E_L(f_\bullet \alpha)$, and Proposition 2.3(ii) applies. $\square$

► *One sentence: the square is an equality, not an isomorphism, because an order-preserving refinement is determined by its source and target (Proposition 3.6), so transporting the two legs of a cell involves no choice. The tree states it as an isomorphism pinned at the identity, which is this equality read through `eqToIso`.*

7 The point

Since Z is terminal, every serial wedge has exactly one bipointed map to it and every map of wedges commutes over it, so $\text{Ch}(Z)$ is the category of shapes and maps of wedges. Write Z_N for the run of Z on N events. Both $\perp_e$ and $\top_e$ have source a run on $|e|$ events, so every cell of Z is a loop at $Z_{|e|}$.

Definition 7.1. For a shape a on N events put $\text{cut}(a) := \{k \in [N-1] : \text{blk}_a(k) = \text{blk}_a(k+1)\}$, and write $\text{cut}(d)$ for the cuts of a chain's shape. For $i \neq k$ in $[N-1]$ let $\text{cox}(i, k) := 3$ if $|i - k| = 1$ and $\text{cox}(i, k) := 2$ otherwise.

Lemma 7.2. $a \mapsto \text{cut}(a)$ is a bijection from the shapes on N events onto the subsets of $[N-1]$, with $|\text{cut}(a)| = \deg a$. Hence $\text{Cell}_n(Z_N, Z_N)$ is the set of n -element subsets of $[N-1]$; write σ_k for the 1-cell with $\text{cut} = \{k\}$. Moreover a chain e of Z with $\deg e = 2$ and $\text{cut}(e) = \{i < k\}$ has $|\text{Sq}(e)| = \text{cox}(i, k)$.

Proof. A shape on N events is a composition of N and $\text{cut}(a)$ is the complement of its set of block boundaries in $[N-1]$, so the first claim is the classical correspondence [Sta12, §1.2] and $|\text{cut}(a)| = N - r = \deg a$. For the last claim, e has a block of dimension 3 exactly when three consecutive events share a block, that is exactly when $k = i + 1$; then $|\text{Sq}(e)| = 3$, and otherwise $|\text{Sq}(e)| = 2$, by Proposition 4.13. $\square$

Lemma 7.3. Let d be a chain on N events, $x \in \mathcal{R}(d)$, and s a step of x . Let $\text{pos}_x(s)$ be the earlier of the two times at which x fires the events of s , so x passes s between times $\text{pos}_x(s)$ and $\text{pos}_x(s) + 1$. Then:

1. pos_x is a bijection $\text{St}(x) \rightarrow \text{cut}(d)$;
2. $\text{pos}_{x^s}(s) = \text{pos}_x(s)$;
3. $\text{pos}_{\perp_d}(s) = p$, where $s = (p, q)$.

If d is a chain of Z , then a cover $x \triangleleft y$ with step s has $\gamma(x \triangleleft y) = \sigma_{\text{pos}_x(s)}$.

Proof. (i) The fibres of blk on the firing order are consecutive intervals of lengths $a_1, \dots, a_r$ (Lemma 4.2), independent of x , so the set of times t with t and $t + 1$ in one block is $\text{cut}(d)$ for every x . A step is a cover of the firing order inside a block (Lemma 4.5), so pos_x lands in $\text{cut}(d)$, and it is injective because distinct covers of a chain have distinct lower ends; both sides have $\deg d$

elements (Lemma 4.5, Lemma 7.2). (ii) Flipping s exchanges its two times and changes no other. (iii) $\perp_d$ fires the events in their own order.

For the last claim, $\gamma(x \leq y) = m_{\{s\}}$, whose blocks are the parts of the firing order cut at every cover but s (Proposition 4.7); so $m_{\{s\}}$ has one block of dimension 2, at times $\text{pos}_x(s)$ and $\text{pos}_x(s) + 1$, and edges elsewhere, so its cut set is $\{\text{pos}_x(s)\}$. $\square$

Proposition 7.4. *Let e be a chain of Z with $\deg e = 2$ and $\text{cut}(e) = \{i < k\}$, and let $\perp_e = x_0 \leq \dots \leq x_{|\text{Sq}(e)|} = \top_e$ be the chain of Lemma 4.11 for the lexicographic order, with s_t the $(t+1)$ -st square. Then $\text{pos}_{x_t}(s_t) = i$ for t even and k for t odd, so*

$$\text{src}(e) = \sigma_i \sigma_k \sigma_i \cdots, \quad \text{tgt}(e) = \sigma_k \sigma_i \sigma_k \cdots,$$

both of length $\text{cox}(i, k)$.

Proof. Every x_t has exactly two steps, passed at i and at k (Lemma 7.3(i)).

For $t = 0$: $\perp_e$ passes (p, q) at p (Lemma 7.3(iii)), and the lexicographic order sorts $\text{Sq}(e)$ by the smaller event first. If e is one block of dimension 3 on events $i, i+1, i+2$, its squares in order are $(i, i+1)$, $(i, i+2)$, $(i+1, i+2)$, with smaller events $i, i, i+1 = k$; if e is two blocks of dimension 2 on $\{i, i+1\}$ and $\{k, k+1\}$, they are $(i, i+1)$ and $(k, k+1)$, with smaller events i and k . Either way the least square is passed at i and the greatest at k .

For the step, s_t and s_{t+1} are both steps of x_{t+1} : the first because a flip of a step is a step, the second by Lemma 4.11. Now $\text{pos}_{x_{t+1}}(s_t) = \text{pos}_{x_t}(s_t)$ (Lemma 7.3(ii)), so $\text{pos}_{x_{t+1}}(s_{t+1})$ is whichever of i and k that is not. The words follow by the last clause of Lemma 7.3, and their length is $|\text{Sq}(e)| = \text{cox}(i, k)$ (Lemma 7.2). The reverse order has the lexicographic order's greatest square first, passed at k , and the same induction applies. $\square$

Definition 7.5. **Artin's braid polygraph** $\mathcal{A}$ has one 0-cell for each $N \in \mathbb{N}$; at N , the loops $\sigma_1, \dots, \sigma_{N-1}$; and, for each pair $i < k$ in $[N-1]$, one 2-cell with words $\sigma_i \sigma_k \sigma_i \cdots$ and $\sigma_k \sigma_i \sigma_k \cdots$, both of length $\text{cox}(i, k)$.

► *One or two sentences: unwind Definition 7.5 to Artin's relations [Art47], so $\langle \mathcal{A} \rangle$ has endomorphism monoid $(B_N^+)^{\text{op}}$ at N and no arrows between distinct objects, i.e. $\langle \mathcal{A} \rangle = \mathbf{Braid}^{+, \text{op}}$.*

Theorem 7.6. $\text{Art}(Z) \cong \mathcal{A}$ in **Poly**.

Proof. The runs of Z are the Z_N and every cell of Z is a loop, so both polygraphs are disjoint unions over N and it suffices to compare at each N . There the 1-cells are indexed by $[N-1]$ and the 2-cells by the pairs $i < k$ (Lemma 7.2), and by Proposition 7.4 the words of the 2-cell at $\{i < k\}$ are those of Definition 7.5. The resulting morphism of polygraphs is bijective in every dimension, hence an isomorphism (Proposition 2.3(iii)). $\square$

Corollary 7.7. $\text{Br}(Z) \simeq \mathbf{Braid}^{+, \text{op}}$.

Proof. Theorem 5.5 at $K = Z$ and Theorem 7.6. $\square$

► *Closing paragraph: combining with Theorem 6.1, the terminal map gives $\text{Br}(!_K) : \text{Br}(K) \rightarrow \text{Br}(Z)$, computed cell by cell — a degree-one chain goes to σ_{pos} for the position at which its run passes its one step, a degree-two chain to the relation of its two positions. Say what this sharpens in [PZ21]. Note that Matsumoto’s theorem plays no role, B_N^+ being taken here as Artin’s presentation.*

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